```
mysql> CREATE DATABASE Hospital_Cursor_Trigger_Lab;
Query OK, 1 row affected (0.01 sec)

mysql> USE Hospital_Cursor_Trigger_Lab;
Database changed
mysql> CREATE TABLE Doctors (DoctorID INT PRIMARY KEY, DoctorName VARCHAR(50),
Department VARCHAR(40), ConsultationFee DECIMAL(10,2), Status VARCHAR(20));
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Doctors VALUES (201,'Dr. Anil','Cardiology',1200,'Active'),(202,'Dr.
Beena','Pediatrics',900,'Active'),(203,'Dr. Chirag','Neurology',1500,'Active'),(204,'Dr.
Divya','Dermatology',1000,'Active'),(205,'Dr. Eshan','Cardiology',1300,'Active'),(206,'Dr.
Farah','Pediatrics',850,'Active');
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE Doctor_Audit (AuditID INT AUTO_INCREMENT PRIMARY KEY, DoctorID
INT, ActionType VARCHAR(30), OldFee DECIMAL(10,2), NewFee DECIMAL(10,2), ActionTime
TIMESTAMP DEFAULT CURRENT_TIMESTAMP);
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE Fee_Log (LogID INT AUTO_INCREMENT PRIMARY KEY, DoctorID INT,
DoctorName VARCHAR(50), OldFee DECIMAL(10,2), NewFee DECIMAL(10,2), ChangedAt
TIMESTAMP DEFAULT CURRENT_TIMESTAMP);
Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE ShowDoctors()
    -> BEGIN
    -> DECLARE done INT DEFAULT FALSE;
```

```
    -> DECLARE v_id INT;
    -> DECLARE v_name VARCHAR(50);
    -> DECLARE cur CURSOR FOR SELECT DoctorID, DoctorName FROM Doctors;
    -> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;
    -> OPEN cur;
    -> read_loop: LOOP
    -> FETCH cur INTO v_id, v_name;
    -> IF done THEN LEAVE read_loop; END IF;
    -> SELECT v_id AS DoctorID, v_name AS DoctorName;
    -> END LOOP;
    -> CLOSE cur;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL ShowDoctors();
+----------+------------+
| DoctorID | DoctorName |
+----------+------------+
|      201 | Dr. Anil   |
+----------+------------+
1 row in set (0.01 sec)

+----------+------------+
| DoctorID | DoctorName |
+----------+------------+
|      202 | Dr. Beena  |
+----------+------------+
```

1 row in set (0.01 sec)

```
+----------+------------+
| DoctorID | DoctorName |
+----------+------------+
|      203 | Dr. Chirag |
+----------+------------+
```

1 row in set (0.01 sec)

```
+----------+------------+
| DoctorID | DoctorName |
+----------+------------+
|      204 | Dr. Divya  |
+----------+------------+
```

1 row in set (0.01 sec)

```
+----------+------------+
| DoctorID | DoctorName |
+----------+------------+
|      205 | Dr. Eshan  |
+----------+------------+
```

1 row in set (0.01 sec)

```
+----------+------------+
| DoctorID | DoctorName |
+----------+------------+
|      206 | Dr. Farah  |
+----------+------------+
```

1 row in set (0.01 sec)

Query OK, 0 rows affected (0.02 sec)

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE DoctorsAboveFee()
    -> BEGIN
    -> DECLARE done INT DEFAULT FALSE;
    -> DECLARE v_id INT;
    -> DECLARE v_name VARCHAR(50);
    -> DECLARE v_fee DECIMAL(10,2);
    -> DECLARE cur CURSOR FOR SELECT DoctorID, DoctorName, ConsultationFee FROM
Doctors;
    -> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;
    -> OPEN cur;
    -> read_loop: LOOP
    -> FETCH cur INTO v_id, v_name, v_fee;
    -> IF done THEN LEAVE read_loop; END IF;
    -> IF v_fee > 1000 THEN SELECT v_id AS DoctorID, v_name AS DoctorName, v_fee AS
ConsultationFee; END IF;
    -> END LOOP;
    -> CLOSE cur;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL DoctorsAboveFee();
+----------+------------+-----------------+
| DoctorID | DoctorName | ConsultationFee |
```

```
+----------+------------+-----------------+
|      201 | Dr. Anil   |         1200.00 |
+----------+------------+-----------------+
1 row in set (0.00 sec)

+----------+------------+-----------------+
| DoctorID | DoctorName | ConsultationFee |
+----------+------------+-----------------+
|      203 | Dr. Chirag |         1500.00 |
+----------+------------+-----------------+
1 row in set (0.01 sec)

+----------+------------+-----------------+
| DoctorID | DoctorName | ConsultationFee |
+----------+------------+-----------------+
|      205 | Dr. Eshan  |         1300.00 |
+----------+------------+-----------------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE TotalConsultationFee(OUT total DECIMAL(10,2))
    -> BEGIN
    -> DECLARE done INT DEFAULT FALSE;
    -> DECLARE v_fee DECIMAL(10,2);
    -> DECLARE cur CURSOR FOR SELECT ConsultationFee FROM Doctors;
    -> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;
```

```
    -> SET total = 0;
    -> OPEN cur;
    -> read_loop: LOOP
    -> FETCH cur INTO v_fee;
    -> IF done THEN LEAVE read_loop; END IF;
    -> SET total = total + v_fee;
    -> END LOOP;
    -> CLOSE cur;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL TotalConsultationFee(@total);
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT @total AS TotalConsultationFee;
+----------------------+
| TotalConsultationFee |
+----------------------+
|              6750.00 |
+----------------------+
1 row in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE CountCardiology(OUT total INT)
    -> BEGIN
    -> DECLARE done INT DEFAULT FALSE;
    -> DECLARE v_dept VARCHAR(40);
```

```
    -> DECLARE cur CURSOR FOR SELECT Department FROM Doctors;
    -> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;
    -> SET total = 0;
    -> OPEN cur;
    -> read_loop: LOOP
    -> FETCH cur INTO v_dept;
    -> IF done THEN LEAVE read_loop; END IF;
    -> IF v_dept = 'Cardiology' THEN SET total = total + 1; END IF;
    -> END LOOP;
    -> CLOSE cur;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL CountCardiology(@total);
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT @total AS CardiologyDoctors;
+------------------+
| CardiologyDoctors |
+------------------+
|                2 |
+------------------+
1 row in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE AverageConsultationFee(OUT average_fee DECIMAL(10,2))
    -> BEGIN
```

```
    -> DECLARE done INT DEFAULT FALSE;
    -> DECLARE v_fee DECIMAL(10,2);
    -> DECLARE total DECIMAL(10,2) DEFAULT 0;
    -> DECLARE cnt INT DEFAULT 0;
    -> DECLARE cur CURSOR FOR SELECT ConsultationFee FROM Doctors;
    -> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;
    -> OPEN cur;
    -> read_loop: LOOP
    -> FETCH cur INTO v_fee;
    -> IF done THEN LEAVE read_loop; END IF;
    -> SET total = total + v_fee;
    -> SET cnt = cnt + 1;
    -> END LOOP;
    -> CLOSE cur;
    -> IF cnt = 0 THEN SET average_fee = 0; ELSE SET average_fee = total / cnt; END IF;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL AverageConsultationFee(@avg);
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT @avg AS AverageConsultationFee;
+------------------------+
| AverageConsultationFee |
+------------------------+
|                1125.00 |
+------------------------+
```

```
1 row in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE IncreasePediatricsFee()
    -> BEGIN
    -> DECLARE done INT DEFAULT FALSE;
    -> DECLARE v_id INT;
    -> DECLARE v_dept VARCHAR(40);
    -> DECLARE cur CURSOR FOR SELECT DoctorID, Department FROM Doctors;
    -> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;
    -> OPEN cur;
    -> read_loop: LOOP
    -> FETCH cur INTO v_id, v_dept;
    -> IF done THEN LEAVE read_loop; END IF;
    -> IF v_dept = 'Pediatrics' THEN UPDATE Doctors SET ConsultationFee = ConsultationFee *
1.10 WHERE DoctorID = v_id; END IF;
    -> END LOOP;
    -> CLOSE cur;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL IncreasePediatricsFee();
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM Doctors;
+----------+------------+-------------+-----------------+--------+
| DoctorID | DoctorName | Department  | ConsultationFee | Status |
```

```
+----------+------------+-------------+-----------------+--------+
|      201 | Dr. Anil   | Cardiology  |         1200.00 | Active |
|      202 | Dr. Beena  | Pediatrics  |          990.00 | Active |
|      203 | Dr. Chirag | Neurology   |         1500.00 | Active |
|      204 | Dr. Divya  | Dermatology |         1000.00 | Active |
|      205 | Dr. Eshan  | Cardiology  |         1300.00 | Active |
|      206 | Dr. Farah  | Pediatrics  |          935.00 | Active |
+----------+------------+-------------+-----------------+--------+
6 rows in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE ReviewLowFeeDoctors()
    -> BEGIN
    -> DECLARE done INT DEFAULT FALSE;
    -> DECLARE v_id INT;
    -> DECLARE v_fee DECIMAL(10,2);
    -> DECLARE cur CURSOR FOR SELECT DoctorID, ConsultationFee FROM Doctors;
    -> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;
    -> OPEN cur;
    -> read_loop: LOOP
    -> FETCH cur INTO v_id, v_fee;
    -> IF done THEN LEAVE read_loop; END IF;
    -> IF v_fee < 1000 THEN UPDATE Doctors SET Status = 'Review' WHERE DoctorID = v_id;
END IF;
    -> END LOOP;
    -> CLOSE cur;
    -> END //
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> DELIMITER ;
mysql> CALL ReviewLowFeeDoctors();
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM Doctors;
+----------+------------+-------------+-----------------+--------+
| DoctorID | DoctorName | Department  | ConsultationFee | Status |
+----------+------------+-------------+-----------------+--------+
|      201 | Dr. Anil   | Cardiology  |         1200.00 | Active |
|      202 | Dr. Beena  | Pediatrics  |          990.00 | Review |
|      203 | Dr. Chirag | Neurology   |         1500.00 | Active |
|      204 | Dr. Divya  | Dermatology |         1000.00 | Active |
|      205 | Dr. Eshan  | Cardiology  |         1300.00 | Active |
|      206 | Dr. Farah  | Pediatrics  |          935.00 | Review |
+----------+------------+-------------+-----------------+--------+
6 rows in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE DoctorReport()
    -> BEGIN
    -> DECLARE done INT DEFAULT FALSE;
    -> DECLARE v_name VARCHAR(50);
    -> DECLARE v_dept VARCHAR(40);
    -> DECLARE v_fee DECIMAL(10,2);
    -> DECLARE cur CURSOR FOR SELECT DoctorName, Department, ConsultationFee FROM
Doctors ORDER BY Department, DoctorName;
    -> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;
```

```
    -> OPEN cur;
    -> read_loop: LOOP
    -> FETCH cur INTO v_name, v_dept, v_fee;
    -> IF done THEN LEAVE read_loop; END IF;
    -> SELECT v_name AS DoctorName, v_dept AS Department, v_fee AS ConsultationFee;
    -> END LOOP;
    -> CLOSE cur;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL DoctorReport();
+------------+------------+-----------------+
| DoctorName | Department | ConsultationFee |
+------------+------------+-----------------+
| Dr. Anil   | Cardiology |         1200.00 |
+------------+------------+-----------------+
1 row in set (0.00 sec)

+------------+------------+-----------------+
| DoctorName | Department | ConsultationFee |
+------------+------------+-----------------+
| Dr. Eshan  | Cardiology |         1300.00 |
+------------+------------+-----------------+
1 row in set (0.01 sec)

+------------+-------------+-----------------+
| DoctorName | Department  | ConsultationFee |
```

```
+------------+-------------+-----------------+
| Dr. Divya  | Dermatology |         1000.00 |
+------------+-------------+-----------------+
1 row in set (0.01 sec)

+------------+------------+-----------------+
| DoctorName | Department | ConsultationFee |
+------------+------------+-----------------+
| Dr. Chirag | Neurology  |         1500.00 |
+------------+------------+-----------------+
1 row in set (0.01 sec)

+------------+------------+-----------------+
| DoctorName | Department | ConsultationFee |
+------------+------------+-----------------+
| Dr. Beena  | Pediatrics |          990.00 |
+------------+------------+-----------------+
1 row in set (0.01 sec)

+------------+------------+-----------------+
| DoctorName | Department | ConsultationFee |
+------------+------------+-----------------+
| Dr. Farah  | Pediatrics |          935.00 |
+------------+------------+-----------------+
1 row in set (0.02 sec)

Query OK, 0 rows affected (0.02 sec)
```

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE CountActiveDoctors(OUT total INT)
    -> BEGIN
    -> DECLARE done INT DEFAULT FALSE;
    -> DECLARE v_status VARCHAR(20);
    -> DECLARE cur CURSOR FOR SELECT Status FROM Doctors;
    -> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;
    -> SET total = 0;
    -> OPEN cur;
    -> read_loop: LOOP
    -> FETCH cur INTO v_status;
    -> IF done THEN LEAVE read_loop; END IF;
    -> IF v_status = 'Active' THEN SET total = total + 1; END IF;
    -> END LOOP;
    -> CLOSE cur;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL CountActiveDoctors(@total);
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT @total AS ActiveDoctors;
+---------------+
| ActiveDoctors |
+---------------+
|             4 |
+---------------+
```

```
1 row in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE IncreaseDoctorFee()
    -> BEGIN
    -> DECLARE done INT DEFAULT FALSE;
    -> DECLARE v_id INT;
    -> DECLARE v_fee DECIMAL(10,2);
    -> DECLARE cur CURSOR FOR SELECT DoctorID, ConsultationFee FROM Doctors;
    -> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;
    -> OPEN cur;
    -> read_loop: LOOP
    -> FETCH cur INTO v_id, v_fee;
    -> IF done THEN LEAVE read_loop; END IF;
    -> IF v_fee >= 1200 THEN UPDATE Doctors SET ConsultationFee = ConsultationFee * 1.05
WHERE DoctorID = v_id; END IF;
    -> END LOOP;
    -> CLOSE cur;
    -> END //
Query OK, 0 rows affected (0.00 sec)

mysql> DELIMITER ;
mysql> CALL IncreaseDoctorFee();
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM Doctors;
+----------+------------+-------------+-----------------+--------+
| DoctorID | DoctorName | Department  | ConsultationFee | Status |
```

```
+----------+------------+-------------+-----------------+--------+
|      201 | Dr. Anil   | Cardiology  |         1260.00 | Active |
|      202 | Dr. Beena  | Pediatrics  |          990.00 | Review |
|      203 | Dr. Chirag | Neurology   |         1575.00 | Active |
|      204 | Dr. Divya  | Dermatology |         1000.00 | Active |
|      205 | Dr. Eshan  | Cardiology  |         1365.00 | Active |
|      206 | Dr. Farah  | Pediatrics  |          935.00 | Review |
+----------+------------+-------------+-----------------+--------+
6 rows in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE TRIGGER AfterDoctorInsert AFTER INSERT ON Doctors FOR EACH ROW BEGIN
INSERT INTO Doctor_Audit (DoctorID, ActionType, NewFee) VALUES (NEW.DoctorID,
'INSERT', NEW.ConsultationFee); END //
Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER ;
mysql> INSERT INTO Doctors VALUES (207,'Dr. Gaurav','Cardiology',1100,'Active');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM Doctor_Audit;
+---------+----------+------------+--------+---------+---------------------+
| AuditID | DoctorID | ActionType | OldFee | NewFee  | ActionTime          |
+---------+----------+------------+--------+---------+---------------------+
|       1 |      207 | INSERT     |   NULL | 1100.00 | 2026-11-01 23:28:15 |
+---------+----------+------------+--------+---------+---------------------+
1 row in set (0.00 sec)

mysql> DELIMITER //
```

```
mysql> CREATE TRIGGER AfterDoctorUpdate AFTER UPDATE ON Doctors FOR EACH ROW BEGIN IF OLD.ConsultationFee <> NEW.ConsultationFee THEN INSERT INTO Doctor_Audit (DoctorID, ActionType, OldFee, NewFee) VALUES (NEW.DoctorID, 'UPDATE', OLD.ConsultationFee, NEW.ConsultationFee); END IF; END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> UPDATE Doctors SET ConsultationFee = 1400 WHERE DoctorID = 201;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM Doctor_Audit;
+---------+----------+------------+---------+---------+---------------------+
| AuditID | DoctorID | ActionType | OldFee  | NewFee  | ActionTime          |
+---------+----------+------------+---------+---------+---------------------+
|       1 |      207 | INSERT     |    NULL | 1100.00 | 2026-11-01 23:28:15 |
|       2 |      201 | UPDATE     | 1260.00 | 1400.00 | 2026-11-01 23:29:07 |
+---------+----------+------------+---------+---------+---------------------+
2 rows in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE TRIGGER UppercaseDoctorName BEFORE INSERT ON Doctors FOR EACH ROW BEGIN SET NEW.DoctorName = UPPER(NEW.DoctorName); END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> INSERT INTO Doctors VALUES (208,'dr. gaurav','Neurology',1000,'Active');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT * FROM Doctors WHERE DoctorID = 208;
+----------+------------+------------+-----------------+--------+
| DoctorID | DoctorName | Department | ConsultationFee | Status |
+----------+------------+------------+-----------------+--------+
|      208 | DR. GAURAV | Neurology  |         1000.00 | Active |
+----------+------------+------------+-----------------+--------+
1 row in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE TRIGGER PreventNegativeFee BEFORE INSERT ON Doctors FOR EACH ROW
BEGIN IF NEW.ConsultationFee < 0 THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT =
'Consultation fee cannot be negative'; END IF; END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> INSERT INTO Doctors VALUES (209,'Dr. Test','Cardiology',-500,'Active');
ERROR 1644 (45000): Consultation fee cannot be negative
mysql> DELIMITER //
mysql> CREATE TRIGGER AfterDoctorDelete AFTER DELETE ON Doctors FOR EACH ROW
BEGIN INSERT INTO Doctor_Audit (DoctorID, ActionType, OldFee) VALUES (OLD.DoctorID,
'DELETE', OLD.ConsultationFee); END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> DELETE FROM Doctors WHERE DoctorID = 207;
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM Doctor_Audit;
+---------+----------+------------+---------+---------+---------------------+
```

| AuditID | DoctorID | ActionType | OldFee | NewFee | ActionTime |

+---------+----------+------------+---------+---------+---------------------+

| 1 | 207 | INSERT | NULL | 1100.00 | 2026-11-01 23:28:15 |

| 2 | 201 | UPDATE | 1260.00 | 1400.00 | 2026-11-01 23:29:07 |

| 3 | 208 | INSERT | NULL | 1000.00 | 2026-11-01 23:30:39 |

| 4 | 207 | DELETE | 1100.00 | NULL | 2026-11-01 23:32:17 |

+---------+----------+------------+---------+---------+---------------------+

4 rows in set (0.00 sec)

mysql> DELIMITER //

mysql> CREATE TRIGGER LogFeeChanges AFTER UPDATE ON Doctors FOR EACH ROW BEGIN IF OLD.ConsultationFee <> NEW.ConsultationFee THEN INSERT INTO Fee_Log (DoctorID, DoctorName, OldFee, NewFee) VALUES (NEW.DoctorID, NEW.DoctorName, OLD.ConsultationFee, NEW.ConsultationFee); END IF; END //

Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;

mysql> UPDATE Doctors SET ConsultationFee = 1600 WHERE DoctorID = 203;

Query OK, 1 row affected (0.01 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM Fee_Log;

+-------+----------+------------+---------+---------+---------------------+

| LogID | DoctorID | DoctorName | OldFee | NewFee | ChangedAt |

+-------+----------+------------+---------+---------+---------------------+

| 1 | 203 | Dr. Chirag | 1575.00 | 1600.00 | 2026-11-01 23:32:57 |

+-------+----------+------------+---------+---------+---------------------+

1 row in set (0.00 sec)

```
mysql> DELIMITER //

mysql> CREATE TRIGGER PreventNegativeFeeUpdate BEFORE UPDATE ON Doctors FOR EACH ROW BEGIN IF NEW.ConsultationFee < 0 THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Updated consultation fee cannot be negative'; END IF; END //

Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER ;

mysql> UPDATE Doctors SET ConsultationFee = -100 WHERE DoctorID = 202;

ERROR 1644 (45000): Updated consultation fee cannot be negative

mysql> DELIMITER //

mysql> CREATE TRIGGER NormalizeDoctorStatus BEFORE UPDATE ON Doctors FOR EACH ROW BEGIN IF LOWER(NEW.Status) = 'active' THEN SET NEW.Status = 'Active'; END IF; END //

Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER ;

mysql> UPDATE Doctors SET Status = 'active' WHERE DoctorID = 204;

Query OK, 0 rows affected (0.01 sec)

Rows matched: 1  Changed: 0  Warnings: 0

mysql> SELECT DoctorID, DoctorName, Status FROM Doctors WHERE DoctorID = 204;

+----------+------------+--------+

| DoctorID | DoctorName | Status |

+----------+------------+--------+

|      204 | Dr. Divya  | Active |

+----------+------------+--------+

1 row in set (0.00 sec)

mysql> DELIMITER //
```

```
mysql> CREATE TRIGGER AuditDepartmentChange AFTER UPDATE ON Doctors FOR EACH ROW BEGIN IF OLD.Department <> NEW.Department THEN INSERT INTO Doctor_Audit (DoctorID, ActionType) VALUES (NEW.DoctorID, 'DEPARTMENT_UPDATE'); END IF; END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> UPDATE Doctors SET Department = 'Cardiology' WHERE DoctorID = 206;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM Doctor_Audit;
+---------+----------+------------------+---------+---------+---------------------+
| AuditID | DoctorID | ActionType       | OldFee  | NewFee  | ActionTime          |
+---------+----------+------------------+---------+---------+---------------------+
|       1 |      207 | INSERT           |    NULL | 1100.00 | 2026-11-01 23:28:15 |
|       2 |      201 | UPDATE           | 1260.00 | 1400.00 | 2026-11-01 23:29:07 |
|       3 |      208 | INSERT           |    NULL | 1000.00 | 2026-11-01 23:30:39 |
|       4 |      207 | DELETE           | 1100.00 |    NULL | 2026-11-01 23:32:17 |
|       5 |      203 | UPDATE           | 1575.00 | 1600.00 | 2026-11-01 23:32:57 |
|       6 |      206 | DEPARTMENT_UPDATE |    NULL |    NULL | 2026-11-01 23:36:09 |
+---------+----------+------------------+---------+---------+---------------------+
6 rows in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE TRIGGER ValidateDoctorStatus BEFORE INSERT ON Doctors FOR EACH ROW BEGIN IF NEW.Status NOT IN ('Active','Review') THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Invalid doctor status'; END IF; END //
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> DELIMITER ;

mysql> INSERT INTO Doctors VALUES (210,'Dr. Karan','Neurology',1200,'Inactive');

ERROR 1644 (45000): Invalid doctor status

mysql> SHOW TRIGGERS;

+-------------------------+--------+---------+-----------------------------
```

| AfterDoctorInsert | INSERT | doctors | BEGIN INSERT INTO Doctor_Audit (DoctorID, ActionType, NewFee) VALUES (NEW.DoctorID, 'INSERT', NEW.ConsultationFee); END | AFTER | 2026-11-01 23:27:36.91 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION | root@localhost | cp850 | cp850_general_ci | utf8mb4_0900_ai_ci |

| PreventNegativeFeeUpdate | UPDATE | doctors | BEGIN IF NEW.ConsultationFee < 0 THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Updated consultation fee cannot be negative'; END IF; END | BEFORE | 2026-11-01 23:34:40.72 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION | root@localhost | cp850 | cp850_general_ci | utf8mb4_0900_ai_ci |

| NormalizeDoctorStatus | UPDATE | doctors | BEGIN IF LOWER(NEW.Status) = 'active' THEN SET NEW.Status = 'Active'; END IF; END | BEFORE | 2026-11-01 23:35:13.30 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION | root@localhost | cp850 | cp850_general_ci | utf8mb4_0900_ai_ci |

| AfterDoctorUpdate | UPDATE | doctors | BEGIN IF OLD.ConsultationFee <> NEW.ConsultationFee THEN INSERT INTO Doctor_Audit (DoctorID, ActionType, OldFee, NewFee) VALUES (NEW.DoctorID, 'UPDATE', OLD.ConsultationFee, NEW.ConsultationFee); END IF; END | AFTER | 2026-11-01 23:28:47.99 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION | root@localhost | cp850 | cp850_general_ci | utf8mb4_0900_ai_ci |

| LogFeeChanges | UPDATE | doctors | BEGIN IF OLD.ConsultationFee <> NEW.ConsultationFee THEN INSERT INTO Fee_Log (DoctorID, DoctorName, OldFee, NewFee) VALUES (NEW.DoctorID, NEW.DoctorName, OLD.ConsultationFee, NEW.ConsultationFee); END IF; END | AFTER | 2026-11-01 23:32:41.72 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION | root@localhost | cp850 | cp850_general_ci | utf8mb4_0900_ai_ci |

| AuditDepartmentChange | UPDATE | doctors | BEGIN IF OLD.Department <> NEW.Department THEN INSERT INTO Doctor_Audit (DoctorID, ActionType) VALUES (NEW.DoctorID, 'DEPARTMENT_UPDATE'); END IF; END | AFTER | 2026-11-01 23:35:53.39 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_

FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION | root@localhost | cp850 | cp850_general_ci | utf8mb4_0900_ai_ci |

| AfterDoctorDelete | DELETE | doctors | BEGIN INSERT INTO Doctor_Audit (DoctorID, ActionType, OldFee) VALUES (OLD.DoctorID, 'DELETE', OLD.ConsultationFee); END | AFTER | 2026-11-01 23:32:00.63 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION | root@localhost | cp850 | cp850_general_ci | utf8mb4_0900_ai_ci |

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```
| AuditID | DoctorID | ActionType       | OldFee  | NewFee  | ActionTime          |
+---------+----------+------------------+---------+---------+---------------------+
|       1 |      207 | INSERT           |    NULL | 1100.00 | 2026-11-01 23:28:15 |
|       2 |      201 | UPDATE           | 1260.00 | 1400.00 | 2026-11-01 23:29:07 |
|       3 |      208 | INSERT           |    NULL | 1000.00 | 2026-11-01 23:30:39 |
|       4 |      207 | DELETE           | 1100.00 |    NULL | 2026-11-01 23:32:17 |
|       5 |      203 | UPDATE           | 1575.00 | 1600.00 | 2026-11-01 23:32:57 |
|       6 |      206 | DEPARTMENT_UPDATE |    NULL |    NULL | 2026-11-01 23:36:09 |
+---------+----------+------------------+---------+---------+---------------------+
6 rows in set (0.00 sec)

mysql> SELECT * FROM Fee_Log;
+-------+----------+------------+---------+---------+---------------------+
| LogID | DoctorID | DoctorName | OldFee  | NewFee  | ChangedAt           |
+-------+----------+------------+---------+---------+---------------------+
|     1 |      203 | Dr. Chirag | 1575.00 | 1600.00 | 2026-11-01 23:32:57 |
+-------+----------+------------+---------+---------+---------------------+
1 row in set (0.00 sec)
```